Real-World Order of Operations

Teacher's Edition

The Period

MA.8.NSO.1.7 Solve multi-step mathematical and real-world problems involving the order of operations with rational numbers including exponents and radicals.

Must Solve multi-step real-world problems with the order of operations, exponents and radicals, in six steps or fewer, and report the answer with its unit.

Must not No keyword strategy and no CUBES — the state guide names circling numbers and boxing key words as a misconception, not a method. No PEMDAS.

Target I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

MTR **MTR.7.1** — Notes I–III, both Examples and boards 2, 3, 4, 6, 7 and 9 · **MTR.2.1** — Notes II and board 9 · **MTR.4.1** — Board 6

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2–4	6 min
Notes	5–7	12 min
Example 1	8–9	3 min
Your Turn	10–11	3 min
Example 2	12–13	3 min
Your Turn	14–15	3 min
Whiteboards	16–33	11 min
Independent Set	34	6 min
IXL	35	5 min
Total		**53 min**

Say these three things out loud today

1. Every substituted value goes into parentheses. That one habit prevents half of what goes wrong today.
2. There is no word that tells you which operation to use. Read what is happening, not which words are there.
3. Round once, at the very end, and write the unit. An answer with no unit is not an answer.

Slide by Slide

1. **Real-World Order of Operations.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite and it carries the day's one habit. *1 retrieves 4.06 — the exponent attaches to its own base; the parentheses decide what that base is (yesterday). 2 retrieves 4.05 — finish each piece, then combine: 12 + 1/2 (last week). 3 retrieves 3.05 — a negative exponent flips the base (last unit). 4 retrieves MA.6.AR.1.3 — substitute, then follow the structure: $2(3)^2 = 2(9)$ (prior grade — the prerequisite this lesson needs).*
3. **Warm-Up — Answers.** Reveal. 1. 0 2. 25/2 3. 4/3 4. 18
4. **Vocabulary — The new words, before they are used.** Read each word and its meaning once; the notes use them next.
5. **Notes — Copy both lines.** Every substituted value goes into parentheses. That single habit prevents half of what goes wrong in this lesson.
6. **Notes — Copy both problems.** There is no word that tells you which operation to use. Read what is happening, not which words are present.
7. **Notes.** Round once, at the very end. Rounding in the middle moves the answer, and it moves it further every step you take afterwards.
8. **Example 1 — A formula with a power in it.** SPOKEN ONLY.
9. **Example 1 — Worked.** Substitute in parentheses, square the speed, then multiply.
10. **Your Turn — Same steps, your numbers. Boards up when done.** Same formula, smaller numbers.
11. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
12. **Example 2 — A cost with a rate and a flat fee.** SPOKEN ONLY.
13. **Example 2 — Worked.** The two per-square-foot prices go inside the group that is multiplied by the area; the flat fee stays outside it.
14. **Your Turn — Same steps, your numbers. Boards up when done.** One rate and one flat fee this time.
15. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
16. **Board 1.** Evaluate $3x^2$ when $x = -2$. → ***12*** *−12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]; 36 — the exponent was applied to 3x [8.NSO.1.7: the exponent belongs to its own base]*
17. **Board 1.**
18. **Board 2.** A cyclist has a mass of 70 kilograms and rides at 8 meters per second. Her bicycle cost 480 dollars. Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v. Find her kinetic energy. → ***2,240 joules*** *The price of the bicycle is not needed. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]; 560 — v^2 read as 2v [8.NSO.1.3: an exponent is not a factor]; 280 — the square dropped altogether; 2,240 with no unit — the answer is in joules*
19. **Board 2.**
20. **Board 3.** Renting a kayak costs 17 dollars plus 9 dollars for each hour. Find the cost of 3 hours. → ***44 dollars*** *78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]; 29 — 17 + 9 + 3, every number added [B1G-M 8.NSO.1.7 misconception: keyword hunting]*
21. **Board 3.**

22–23. Board 4. A square floor of side 8 feet is tiled at 3.50 dollars a square foot. Installation is a flat 60 dollars. Find the total cost. → ***284 dollars*** *4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]; 88 — 8(3.50) + 60: the side used instead of the area*

24–25. Board 5. Evaluate ½mv² when m = 40 and v = 3. Which is the value? → ***A — 180*** *(B) v² was read as 2v [8.NSO.1.3: an exponent is not a factor]; (C) the one half was dropped [8.NSO.1.7: every factor in the formula is used]; (D) the exponent was applied to the whole product, (½ · 40 · 3)² [8.NSO.1.7: the exponent belongs to the v alone] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

26–27. Board 6. A square patio of side 12 feet is sealed at 1.25 dollars a square foot. Delivery of the sealer is a flat 38 dollars. Which expression gives the total cost? → ***D — $12^2(1.25) + 38$*** *(A) the side was used where the area is needed [8.NSO.1.7: read what the rate is per]; (B) the prices were added to the area instead of multiplied by it [B1G-M 8.NSO.1.7 misconception: keyword hunting]; (C) the flat delivery charge was put inside the area group, so it is charged on every square foot [8.NSO.1.7: a flat fee is not a rate] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.*

28–29. Board 7. A square garden has 4 sides, each √150 feet long. Find its perimeter, to the nearest hundredth of a foot. → ***48.99 feet*** *48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]; 48.99 with no unit — the answer is in feet*

30–31. Board 8. Evaluate $(a^2 - b) \div c$ when a = 7, b = 4, c = 5. → ***9*** *48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]; 5.8 — 49 ÷ 5 done first, then 4 subtracted*

32–33. Board 9. One pizza order costs 13 dollars plus 2 dollars for each of 5 toppings. Another costs 13 dollars for each of 2 pizzas, plus 5 dollars for delivery. Find both totals, and say why the key words could not have told you. → ***23 dollars and 31 dollars*** *23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]; 23 twice — the second structure was not read*

34. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

35. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Variation. Questions 1–4 substitute into four shapes: a coefficient with a power, a formula with two letters, a fraction bar, and a binomial inside a power. 5 and 6 are cost expressions — one square, one rectangle — where the student has to decide what multiplies the area. 7 and 8 are two real formulas, each evaluated three times, so the structure is fixed and only the numbers move; 7 also requires rounding to a stated place. 9 works backwards in three shapes, ending with a cost equation that has to be unwound. The pattern breaks at 10, where two problems with the same numbers and the same words have different answers. 11 is the diagnostic multiple choice on cost structure and 12 the select-all on what an exponent is attached to. The Additional sheet mirrors each position.

Misconceptions to Watch

Board 1. −12 — the exponent was applied to the 2 and not to the whole substituted value [8.NSO.1.7: substitute in parentheses]

Board 2. anything built from 480 — the unneeded figure was used [the formula takes a mass and a speed]

Board 3. 78 — (17 + 9)(3): the flat 17 was charged three times [8.NSO.1.7: only the hourly price is repeated]

Board 4. 4,064 — 64(3.50 + 60): the flat fee was put inside the area group [8.NSO.1.7: a flat fee is not a rate]

Board 5. v^2 was read as 2v [8.NSO.1.3: an exponent is not a factor]

Board 6. the side was used where the area is needed [8.NSO.1.7: read what the rate is per]

Board 7. 48 — the side was rounded to 12 before multiplying [Unit 2: round once, at the end]

Board 8. 48.2 — only the 4 was divided by 5 [8.NSO.1.7: the bar groups the whole numerator]

Board 9. 23 and 31 with no explanation — the item asks why [same key words, different structures: the explanation is the evidence]

Materials Whiteboards and markers. Calculators are allowed on everything and are the right tool for the BMI and kinetic-energy items; the work being assessed is the expression, not the arithmetic.